Fertility and Housing Tenure Choice: Model Writeup

Tommaso De Santo

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1 Model

The model is a discrete-time lifecycle economy with endogenous fertility, housing tenure choice, and spatial sorting. Households save, choose where to live, decide whether to rent or own, and choose fertility. Children enter preferences and resources through consumption and housing requirements. Housing is local, supplied upward sloping in each location, and costly to access because of down-payment and moving frictions. The mechanism is that children require space, space is more expensive in central locations, and tenure constraints affect family formation over the lifecycle.

The environment combines three standard ingredients. The fertility block follows quantitative lifecycle models in which children are chosen by households and affect both current resources and continuation values (Sommer, 2016; Doepke and Kindermann, 2019). The tenure block follows housing-macro models with collateral constraints and discrete owner-occupied housing (Henderson and Ioannides, 1983; Sommer et al., 2013; Sommer and Sullivan, 2018). The spatial block is Rosen–Roback: households sort across locations with different amenities and prices, and local housing supply determines how demand pressure maps into prices (Rosen, 1979; Roback, 1982). The housing block follows the dynamic urban housing structure in Greaney et al. (2025).

1.1 Environment

Time is discrete. Households enter adult life at age $a = 1$, retire after age J_R[1], and die at the end of age J.

There are two residential locations, $\mathcal{I} = \{P, C\}$, where P denotes the periphery and C denotes the center. A unit of housing in location i has purchase price p_i and rental price r_i. Location i also has amenity ξ_i, housing supply scale H_{0i}, and housing supply elasticity η_i. The amenity ξ_i enters the location choice value, H_{0i} shifts the level of local housing supply, and η_i governs the slope of that supply curve. The center is the high-amenity, high-price location in the benchmark.

The individual state at the beginning of a period is $x = (b, d, i, a, n, s)$. Here b is liquid wealth, d is tenure, i is current location, a is age, n is completed fertility, and s is the child-age state. Tenure belongs to $\mathcal{D} \equiv \{R, O_1, \ldots, O_{N_H}\}$. R denotes renting. O_k denotes ownership of a house

[1] I might consider simplifying the lifecycle structure to just some stages, to improve computational efficiency. The next step is, indeed, clearly determining what are the empirical targets for the model, and so whether I want to match very well lifecycle profiles or not.

with physical size H_k, where $H_1 < \cdots < H_{N_H}$. Owners choose from this discrete ladder[2], while renters choose rental space continuously up to a cap $\bar{h}^R$.

1.2 Preferences

Let $m(n, s)$ be the number of dependent children living at home[3]. Completed fertility n and current dependents $m(n, s)$ differ because children eventually leave the household. Flow utility is Stone–Geary over nondurable consumption and housing services, wrapped in CRRA:

$$u(c, \mathbf{h}; n, s) = \frac{\left[(c - \bar{c}(n, s))^{\alpha} \left(\mathbf{h} - \bar{h}(n, s)\right)^{1-\alpha}\right]^{1-\sigma}}{1 - \sigma} + \psi_{\text{child}} m(n, s), \tag{1}$$

where $\alpha \in (0, 1)$ is the expenditure-share parameter inside the consumption-housing aggregator, $\sigma > 1$ is curvature, and ψ_{child} is the flow value of each dependent child.[4] The Stone–Geary terms $\bar{c}(n, s)$ and $\bar{h}(n, s)$ are minimum consumption and housing needs:

$$\begin{aligned}
\bar{c}(n, s) &= \bar{c}_0 + \bar{c}_n m(n, s), \\
\bar{h}(n, s) &= \bar{h}_0 + \mathbb{I}\{m(n, s) \geq 1\}\bar{h}_{\text{jump}} + \bar{h}_n m(n, s).
\end{aligned}$$

Here $\bar{c}_0$ and $\bar{h}_0$ are baseline needs for a childless household. The parameter $\bar{c}_n$ is the per-child consumption requirement. The parameter $\bar{h}_{\text{jump}}$ is the fixed housing requirement associated with having any dependent child, while $\bar{h}_n$ is the additional housing requirement per dependent child. The first child therefore raises required housing discretely; additional children raise required housing linearly. The local price of space is then part of the cost of children.

Housing services differ by tenure. Let

$$\mathbf{h}(h^O, h^R) = \chi h^O + h^R$$

be the housing services entering utility. Renters have $h^O = 0$ and choose h^R. Owners of rung k have $h^O = H_k$ and $h^R = 0$. The parameter $\chi > 1$ captures the service premium of owner-occupied housing. In market clearing, however, owners demand the physical quantity H_k, not the service quantity χH_k.

At death, households receive warm-glow bequest utility. Let $\psi \in [0, 1)$ denote the proportional transaction cost paid when owner-occupied housing is liquidated. Bequeathable wealth is

$$\widetilde{b} = \begin{cases} b, & d = R, \\ b + (1 - \psi) p_i H_k, & d = O_k. \end{cases}$$

The bequest term is

$$\mathcal{B}(\widetilde{b}, n) = \theta_0 (1 + \theta_n n) \frac{(\theta_1 + \max\{\widetilde{b}, 0\})^{1-\sigma} - \theta_1^{1-\sigma}}{1 - \sigma}. \tag{2}$$

[2]This makes housing more lumpy and makes the model computationally lighter, but might be amended. Some further thinking is required on the housing market structure.

[3]Some notation improvement is probably required, but in any case, the presence of both m and n allows us to distinguish children ever had with children currently at home. which affects the utility.

[4]The position of the ψ_{child} term determines how the child benefit relates to income, so it requires some further thought.

The parameter θ_0 governs the overall strength of the bequest motive. The parameter θ_n scales the bequest motive with completed fertility, so completed fertility affects bequests even after children leave the household. The shift θ_1 controls curvature near zero wealth. The normalization sets bequest utility equal to zero when bequeathable wealth is nonpositive. Parenthood can therefore affect late-life saving and ownership without adding a mechanical terminal-value penalty for having children.

1.3 Firms, Income, and Assets

Competitive firms produce the consumption good with aggregate efficiency labor, $Y = AL$. The consumption good is the numeraire. Competition implies $w = A$, where w is the common wage per efficiency unit. Wages do not vary by location, so spatial sorting is driven by amenities, housing costs, tenure frictions, and moving wedges, not by local wage differences.

Workers supply labor inelastically. A worker of age a supplies e_a efficiency units, so labor income is

$$y_a = (1 - \tau_{\text{pay}}) w e_a, \qquad a \leq J_R,$$

where e_a is the deterministic age-efficiency profile and τ_{pay} is the payroll tax rate. Retired households receive

$$y_a = p^{\text{ret}}, \qquad a > J_R.$$

The pension benefit is determined in equilibrium by the pay-as-you-go government budget.

Given the level distribution g, aggregate efficiency labor supply is

$$L^s(g) = \int \mathbb{I}\{a \leq J_R\} e_a \, dg(x).$$

In equilibrium firms hire this labor supply, so $L = L^s(g)$ and output is $Y = AL$.

Liquid wealth earns gross return $R_b \equiv 1 + q$. A renter cannot borrow:

$$b' \geq 0.$$

At origination, a buyer of house H_k in location i must satisfy

$$b \geq -\phi p_i H_k, \tag{3}$$

where b denotes liquid wealth immediately after the purchase. Here ϕ is the financed share of house value. The required down payment is therefore $(1 - \phi) p_i H_k$. A homeowner who keeps the current house is not forced to delever after a price change, but cannot increase debt while below the origination bound:

$$b' \geq \underline{b}_i^O(b, H_k) \equiv \min\{b, -\phi p_i H_k\}.$$

1.4 Housing and Budgets

Given rental price r_i, a renter chooses c, h^R, and b' subject to

$$c + r_i h^R + b' = R_b b + y_a, \qquad 0 \leq h^R \leq \bar{h}^R, \qquad b' \geq 0.$$

An owner of house H_k chooses c and b' subject to

$$c + \delta p_i H_k + b' = R_b b + y_a, \qquad b' \geq \underline{b}_i^O(b, H_k).$$

Mortgage debt is negative liquid wealth, so mortgage interest is already embedded in $R_b b$. Owners pay maintenance directly.

Housing adjustment is costly. The same transaction-cost parameter ψ used in bequeathable wealth applies when a household sells a house. Selling a house in location i yields net proceeds

$$(1 - \psi) p_i H_k.$$

A moving owner sells the origin house before choosing tenure in the destination. A stayer can keep the current house, sell and rent, or move to another owner rung if the budget and collateral constraints allow it.

1.5 Fertility and Child Aging

During the fertile window

$$a \in \{A_f^{\text{start}}, \dots, A_f^{\text{end}}\},$$

a childless household chooses $n' \in \mathcal{N} \equiv \{0, 1, \dots, \bar{n}\}$. The choice $n' = 0$ means no birth in the current period. A positive choice $n' > 0$ fixes completed fertility forever. A household that reaches the end of the fertile window without a positive fertility choice remains childless. Children are born together and age together through K child stages. The model therefore speaks to completed fertility and additional-child housing demand, not to sequential parity-progression hazards.

Let $s = 0$ denote no children and $s = 1, \dots, K$ denote dependent child stages. After the last dependent stage, children mature and leave the household. Child aging follows

$$\Pi^c(s' | s, n).$$

The matrix Π^c is the transition matrix over child-age states, conditional on completed fertility. In the benchmark parameterization, dependent childhood lasts about 18 years. While children are at home, $m(n, s) = n$. After maturity, $m(n, s) = 0$, so the consumption and housing floors return to their baseline levels, while completed fertility remains payoff-relevant through the bequest motive. Let

$$\lambda_M(s, n) = \Pr_{\Pi^c}\left(s' \notin \{0, 1, \dots, K\} \mid s, n\right)$$

be the probability that dependent children mature next period. The same aging matrix determines the flow of city-born children who become young adults:

$$B(g) = \int n \lambda_M(s, n) dg(x). \tag{4}$$

Mature city-born children are measured in entrant-household equivalents and are counted as potential entrants; the outside option determines whether they enter the modeled city.

1.6 Household Problem

Within the period, the order of decisions is fertility, location, tenure and house size, and then consumption, rental size, and savings. The recursion is written backward in the reverse order. The discount factor is β. Let the terminal value be

$$V_{J+1}(b,d,i,n,s) = \mathcal{B}(\widetilde{b},n),$$

and define the continuation value after child aging by

$$\bar{V}_{a+1}(b',d',i',n,s) = \sum_{s'} \Pi^c(s'|s,n) V_{a+1}(b',d',i',n,s').$$

Conditional on being a renter in location i, the household value is

$$\begin{aligned} V_a^R(b,i,n,s) = \max_{c,h^R,b'} \quad & u(c,\mathbf{h}(0,h^R);n,s) + \beta \bar{V}_{a+1}(b',R,i,n,s) \\ \text{s.t.} \quad & c + r_i h^R + b' = R_b b + y_a, \\ & 0 \le h^R \le \bar{h}^R, \qquad b' \ge 0. \end{aligned} \tag{5}$$

Conditional on owning house H_k in location i, the value is

$$\begin{aligned} V_a^{O_k}(b,i,n,s) = \max_{c,b'} \quad & u(c,\mathbf{h}(H_k,0);n,s) + \beta \bar{V}_{a+1}(b',O_k,i,n,s) \\ \text{s.t.} \quad & c + \delta p_i H_k + b' = R_b b + y_a, \\ & b' \ge \underline{b}_i^O(b,H_k). \end{aligned} \tag{6}$$

Define $H(R) = 0$ and $H(O_k) = H_k$. After choosing destination j, a household with state (b,d,i) carries liquid wealth and owner-occupied housing

$$\widetilde{b}^R(b,d,i;j) = b + \mathbb{I}\{j \neq i\}(1-\psi)p_i H(d), \qquad \widetilde{H}^R(d,i;j) = \mathbb{I}\{j = i\}H(d).$$

If it then chooses tenure d' in destination j, liquid wealth is

$$\widetilde{b}^H(b,d,i;j,d') = \widetilde{b}^R(b,d,i;j) + \mathbb{I}\{H(d') \neq \widetilde{H}^R(d,i;j)\}\left[(1-\psi)p_j \widetilde{H}^R(d,i;j) - p_j H(d')\right]. \tag{7}$$

The feasible tenure set in destination j is

$$\mathcal{D}(b,d,i;j) = \left\{d' \in \mathcal{D} : \big(j = i \text{ and } d' = d\big) \text{ or } \widetilde{b}^H(b,d,i;j,d') \ge \underline{b}_j(d')\right\},$$

where $\underline{b}_j(R) = 0$ and $\underline{b}_j(O_k) = -\phi p_j H_k$. The tenure value in destination j is

$$V_a^T(b,d,i;j,n,s) = \max_{d' \in \mathcal{D}(b,d,i;j)} V_a^{d'}\left(\widetilde{b}^H(b,d,i;j,d'),j,n,s\right). \tag{8}$$

Here $V_a^{d'} = V_a^R$ if $d' = R$ and $V_a^{d'} = V_a^{O_k}$ if $d' = O_k$.

Location choice has additive Type-I extreme-value shocks with scale κ_ℓ. The parameter κ_ℓ governs the amount of residual dispersion in location choices; as κ_ℓ goes to zero, location choice converges to a deterministic maximum. A household currently in location i evaluates destination j using

$$\widetilde{V}_a^L(j \mid \cdot) = V_a^T(b,d,i;j,n,s) + \xi_j - \zeta_{ij},$$

where ζ_{ij} is the bilateral moving wedge, with ζ_{ii} normalized to zero. The destination probabilities are

$$\pi_a^L(j \mid \cdot) = \frac{\exp\left(\widetilde{V}_a^L(j \mid \cdot)/\kappa_\ell\right)}{\sum_{m\in\mathcal{I}} \exp\left(\widetilde{V}_a^L(m \mid \cdot)/\kappa_\ell\right)}, \tag{9}$$

and the location value is

$$V_a^L(b,d,i,n,s) = \kappa_\ell \log \sum_{m\in\mathcal{I}} \exp\left(\widetilde{V}_a^L(m \mid \cdot)/\kappa_\ell\right). \tag{10}$$

Childless households in the fertile window face a stopping problem with additive Type-I extreme-value shocks with scale κ_f. The option $n' = 0$ means delaying fertility for one period. A positive option $n' > 0$ fixes completed fertility permanently. This shock is part of the household fertility problem, not part of the population-entry closure below. Let $s(n') = 0$ if $n' = 0$ and $s(n') = 1$ if $n' > 0$. The value of choosing option $n' \in \mathcal{N}$ is

$$\widetilde{V}_a^F(n' \mid \cdot) = V_a^L(b,d,i,n',s(n')).$$

Fertility probabilities are

$$\pi_a^F(n' \mid \cdot) = \frac{\exp\left(\widetilde{V}_a^F(n' \mid \cdot)/\kappa_f\right)}{\sum_{m\in\mathcal{N}} \exp\left(\widetilde{V}_a^F(m \mid \cdot)/\kappa_f\right)}, \tag{11}$$

and the full household value is

$$V_a(b,d,i,n,s) = \kappa_f \log \sum_{m\in\mathcal{N}} \exp\left(\widetilde{V}_a^F(m \mid \cdot)/\kappa_f\right). \tag{12}$$

Outside the fertility window, or after fertility has already been chosen, this stage is degenerate and $V_a(b,d,i,n,s) = V_a^L(b,d,i,n,s)$.

1.7 Entry, Outside Option, and City Size

With endogenous fertility, stationarity requires specifying how city-born children become adult potential entrants and how the mass of adult households is pinned down. The benchmark closure keeps the city stationary while allowing its scale to be an equilibrium object. Let g denote the level distribution of adult households, and let

$$S = \int dg.$$

Write this distribution as[5]

$$g = S\mu, \qquad \int d\mu = 1,$$

[5]More on this later. I understand this is the main 'technical' issue that emerged in the seminar. It is not fully solved. At the end of this document there is a discussion of the things. I hope to get feedback on this.

where μ is the composition of the city. The normalization is only a computational convenience; market clearing uses g.

There is a pool of potential young-adult entrants. Each period the city receives an exogenous flow $M \geq 0$ of outside-born potential entrants. It also receives the mature city-born flow $B(g)$ generated by the child-aging matrix. Thus the potential-entrant pool is

$$Q(g) = M + B(g). \tag{13}$$

Actual entry from this pool is determined by the outside option below.

A potential entrant enters the modeled city only if the city is sufficiently attractive relative to the outside economy. The value of entering location i is

$$W_i^E(p) = V_1(b^E, R, i, 0, 0; p),$$

where entrants begin as childless renters with liquid wealth b^E. The value $W_i^E(p)$ is computed from the household problem at prices p. The outside value $\bar{W}^E$ is exogenous. Potential entrants draw idiosyncratic Type-I extreme-value tastes over the modeled locations and the outside economy, with scale parameter κ_E. These entry shocks are distinct from the incumbent location shocks governed by κ_ℓ: κ_E governs the city-versus-outside margin, while κ_ℓ governs re-sorting across locations inside the modeled city. The unconditional probability of entering location i is

$$\pi_i^E(p) = \frac{\exp\left(W_i^E(p)/\kappa_E\right)}{\exp\left(\bar{W}^E/\kappa_E\right) + \sum_{m\in\mathcal{I}} \exp\left(W_m^E(p)/\kappa_E\right)}. \tag{14}$$

The outside probability is

$$\pi_0^E(p) = \frac{\exp\left(\bar{W}^E/\kappa_E\right)}{\exp\left(\bar{W}^E/\kappa_E\right) + \sum_{m\in\mathcal{I}} \exp\left(W_m^E(p)/\kappa_E\right)}.$$

The total city-entry probability and conditional entrant location shares are

$$q^E(p) = 1 - \pi_0^E(p) = \sum_{i\in\mathcal{I}} \pi_i^E(p), \qquad \omega_i^E(p) = \frac{\pi_i^E(p)}{q^E(p)}.$$

Entry into location i is

$$E_i(g; p) = Q(g)\pi_i^E(p) = Q(g)q^E(p)\omega_i^E(p). \tag{15}$$

The remaining mass $Q(g)(1 - q^E(p))$ stays in the outside economy.

The outside economy supplies potential entrants and absorbs those who do not enter the modeled city. For a candidate price vector p, household policies imply a stationary composition $\mu(p)$. Let

$$E_0(p) = \int \mathbb{I}\{a = 1\} d\mu(p)$$

be the young-adult entry flow required to reproduce one unit of that composition, and let $B_0(p) = B(\mu(p))$ be the mature city-born flow per unit of city scale. The city scale is pinned down by the stationarity condition

$$SE_0(p) = q^E(p)\left[M + SB_0(p)\right]. \tag{16}$$

If

$$E_0(p) > q^E(p)B_0(p), \tag{17}$$

then the stationary city scale is

$$S(p) = \frac{q^E(p)M}{E_0(p) - q^E(p)B_0(p)}. \tag{18}$$

The denominator is the net replacement gap. If condition (17) fails, the maintained stationary closure does not admit a finite city.[6]

The baseline uses the normalization $S = 1$. Given the baseline objects E_0^*, B_0^*, and $q^{E,*}$, the outside-born flow is set residually from (16):

$$M = \frac{E_0^*}{q^{E,*}} - B_0^*.$$

Thus the benchmark is close to a fixed-mass normalization. The difference is that, in counterfactuals, M and $\bar{W}^E$ are held fixed and (18) determines the new scale $S(p)$.

1.8 Housing Supply and Market Clearing

Housing supply in location i is

$$H_i^S(r_i) = H_{0i} r_i^{\eta_i}. \tag{19}$$

The scale H_{0i} captures reduced-form land costs, regulation, and other supply-side wedges. The elasticity η_i governs the construction response. Thus the supply side has only a level parameter and an elasticity in each location.

Aggregate physical housing demand is

$$H_i^D(g) = \int \left[\mathbb{I}\{d = R, i(x) = i\} h^R(x) + \sum_{k=1}^{N_H} \mathbb{I}\{d = O_k, i(x) = i\} H_k \right] dg(x). \tag{20}$$

If $g = S\mu$, this is

$$H_i^D(g) = S\widehat{H}_i^D(\mu).$$

Market clearing is $H_i^D(g) = H_i^S(r_i)$. The factor S appears only because $g = S\mu$ writes level demand using the normalized composition. In the benchmark, $S = 1$; in counterfactuals, the price vector solves

$$S(p)\widehat{H}_i^D(p) = H_{0i} r_i^{\eta_i}, \qquad i \in \mathcal{I}. \tag{21}$$

[6] With deterministic aging and death after age J, $E_0(p)$ is the stationary mass of age-one households per unit of city scale. Equation (16) says that the required level flow of young adults, $SE_0(p)$, equals the fraction $q^E(p)$ of the potential entrant pool $M + SB_0(p)$ that chooses the modeled city. At a candidate price vector p, the household problem implies policies, $\mu(p)$, $q^E(p)$, and the per-unit objects $E_0(p)$, $B_0(p)$, and $\widehat{H}_i^D(p)$. Equation (18) gives $S(p)$. The equilibrium price vector sets the housing-market residuals $S(p)\widehat{H}_i^D(p) - H_i^S(r_i)$ equal to zero. The finite-scale condition is evaluated at equilibrium prices. Quantitatively, the gap $E_0(p) - q^E(p)B_0(p)$, or equivalently the ratio $q^E(p)B_0(p)/E_0(p)$, measures proximity to the autoreproduction frontier.

House prices satisfy the user-cost relation

$$r_i = (q + \delta)p_i.$$

Thus, for each candidate p, the household problem and entry closure determine $\widehat{H}_i^D(p)$ and $S(p)$; equilibrium rents and house prices are those for which level supply equals level demand in both locations.

2 Stationary Equilibrium

The equilibrium is stationary in prices, policies, and the level distribution of adult households. Stationarity is imposed on g, not only on the normalized composition μ. Entry, aging, fertility, location choice, tenure choice, savings, and death reproduce the same level distribution every period.

Definition 1 (Stationary equilibrium)**.** *A stationary equilibrium is a collection*

$$\mathcal{X}^* = \big(\{p_i, r_i\}_{i\in\mathcal{I}}, w, L, Y, p^{\text{ret}}, \{V_a, V_a^R, V_a^T, V_a^L, \widetilde{V}_a^L, \widetilde{V}_a^F\}_a, \{V_a^{O_k}\}_{a,k}, \{c, h^R, b', d', i', n'\}, \\ \{\pi^L, \pi^F, \pi^E, \pi_0^E, q^E, \omega^E\}, g, \{E_i, H_i^D, H_i^S\}_{i\in\mathcal{I}}\big)$$

such that:

1. ***Household optimization.*** *Given prices, amenities, and moving wedges, value functions solve the household problem in (5)–(6), with tenure choice (8), location probabilities (9), fertility probabilities (11), and terminal bequest utility (2). The policies c, h^R, b', d', i', and n' are associated optimal policies.*

2. ***Entry and outside option.*** *The child-aging matrix implies a mature city-born flow $B(g)$. The potential entrant pool is $Q(g) = M + B(g)$. Entry probabilities satisfy (14), the city entry rate is $q^E(p) = 1 - \pi_0^E(p) = \sum_i \pi_i^E(p)$, and entry masses satisfy*

   $$E_i(g; p) = Q(g)q^E(p)\omega_i^E(p), \qquad i \in \mathcal{I}.$$

   The outside mass is $Q(g)(1 - q^E(p))$.

3. ***Firms and labor market.*** *The common wage satisfies $w = A$. Aggregate efficiency labor supply is*

   $$L^s(g) = \int \mathbb{I}\{a \le J_R\} e_a \, dg(x),$$

   and the labor market clears:

   $$L = L^s(g).$$

   Output is $Y = AL$.

4. ***Housing markets and user cost.*** *In every location,*

   $$H_i^D(g) = H_i^S(r_i), \qquad r_i = (q + \delta)p_i.$$

5. ***Government budget.*** *Payroll-tax revenue finances pension benefits:*

$$\tau_{\text{pay}} w L = p^{\text{ret}} \sum_i \sum_{a > J_R} g_i(a).$$

Here $g_i(a)$ is the marginal mass of households of age a living in location i.

6. ***Stationary distribution.*** *The distribution g is consistent with individual decisions and entry. Starting from g, the optimal policies, fertility and location probabilities, child aging, deterministic aging, death, and entrant masses $\{E_i(g; p)\}_{i \in \mathcal{I}}$ generate the same distribution next period:*

$$g' = \mathcal{T}(g) = g.$$

Entrants enter at $x_i^E = (b^E, R, i, 1, 0, 0)$. Total adult population is $S = \int dg(x)$ and is an equilibrium outcome.

3 Population Closure: Alternatives

The closure above is the benchmark. The alternatives below hold the household problem and housing environment fixed while changing the interpretation of counterfactuals.

Fixed adult mass. The simplest closure sets $\int dg = 1$ and keeps adult population fixed. This is common in lifecycle fertility models and in partial-equilibrium fertility exercises. Fertility then changes the composition of households, but not the mass of households in the city. Housing demand changes because parents want more space, sort differently, and choose tenure differently. It does not change because the city becomes larger. This is close in spirit to exercises that take the adult demographic envelope or prices as given, including Moreno-Maldonado and Santamaria (2024) and Karabarbounis and Nord (2025). The advantage is tractability; the cost is that the scale channel is shut down.

Reduced-form renewal valve. A second possibility is to ignore entry values and keep only demographic accounting:

$$SE_0(p) = M + \rho S B_0(p), \qquad S = \frac{M}{E_0(p) - \rho B_0(p)}.$$

The city receives an outside flow M and retains a fixed fraction ρ of mature city-born children. This delivers a finite stationary scale without introducing a transition path or balanced growth. Fertility affects aggregate housing demand through S. The limitation is that retention is mechanical: city attractiveness affects fertility and housing choices, but it does not directly affect the share of potential entrants who enter the city.

Outside-option entry. The benchmark uses the outside-option version. It keeps the same stationary accounting logic, but makes city entry depend on the value of entering the city relative to an outside economy:

$$SE_0(p) = q^E(p) \left[M + SB_0(p) \right].$$

The equilibrium is still stationary: it solves for a level S, not for a population growth rate. Relative to the reduced-form valve, housing prices discipline city size through entry values. If fertility raises the flow of mature city-born children, the potential-entrant pool rises. If the larger city raises housing prices, entry values fall and the outside option absorbs more of the potential flow. The model then has a scale channel without becoming a theory of city growth.

Full demographic city growth. The most structural alternative is a sequence model in which births today become adult households in future periods, the age distribution evolves over calendar time, and housing markets clear along the transition. This is the natural object for a paper about city growth and fertility dynamics, and it is the direction taken by Coeurdacier et al. (2023). It requires a path for age masses, migration flows, survival, and housing supply. The object here is instead a stationary city in which fertility, housing demand, tenure, and sorting are jointly determined.

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